Chaitanya Kotlapati

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Summary

Backend engineer (4+ years) specializing in Go and Python, building APIs, distributed systems, and high-throughput data pipelines. Co-Founder/Principal Architect at Avalon, owning the Go backend of a zero-config deployment platform: webhook-driven deployments, a pre-deployment validation pipeline, and a tree-sitter based code analysis engine. Previously led a 5-member backend team at SecPod, owning **~80%** of core modules, building a misconfiguration detection engine with **~95% accuracy**, and speeding up parallel data pipelines by **~20%**. Hands-on with Kubernetes, AWS, and Azure. IIT Madras M.Tech.

Skills

- **Languages:** Go, Python, Java, C++
- **Infrastructure:** Kubernetes, AKS, Docker, distributed systems, CI/CD, container orchestration
- **Cloud:** Azure (AKS, Entra ID, VMs), AWS (IAM, EC2, S3, CloudTrail)
- **Backend:** REST APIs, Gin, FastAPI, Flask, asynchronous/parallel processing, microservices
- **Databases/Messaging:** PostgreSQL, MongoDB, MySQL, Redis, AWS SQS, AWS SNS
- **Tools:** Git/SVN, CNAPP concepts, misconfiguration detection

Experience

Co-Founder & Principal Architect — Avalon (Isorropia AI) *May 2026 – Present*

- Building Avalon, a zero-config, cloud-agnostic deployment platform that analyzes a repository and auto-generates Dockerfiles to deploy applications built on any framework.
- Own core backend architecture in Go, including auto-deployment triggered by GitHub webhook events and a pre-deployment validation pipeline with automatic test-suite detection.
- Design and operate the underlying Azure/Kubernetes (AKS) infrastructure: provisioning, ingress, and deployment orchestration.
- Building a **code analysis engine** using tree-sitter based AST parsing to detect frameworks, services, and dependencies across arbitrary repositories.
- Co-founded Avalon as CTO, leading technical direction and **guiding a team of 2 interns**.

Lead Software Engineer, Backend – Cloud Security — SecPod, Bangalore *May 2024 – May 2026*

- Led and built CNAPP features for AWS (IAM, EC2, S3, CloudTrail) and Azure (Entra ID, VMs), owning **~80%** of backend modules and **leading a 5-member team**.
- Built a **misconfiguration detection engine** with **~95% accuracy** that reduced customer security incidents by **~40%**.
- Optimized async/parallel collection pipelines, improving performance by **~20%** and enabling monitoring of **200+ cloud resource types**.
- Engineered a centralized data-feed service for security modules, improving cross-team development productivity by **~35%**.

Software Engineer — SecPod, Bangalore *May 2022 – May 2024*

- Developed a risk prioritization workflow to rank vulnerabilities/misconfigurations using asset and configuration context, with decision-tree based scoring to improve signal quality for downstream modules.
- Built a dynamic endpoint scoring system, exposing scores to UI and reporting components through RESTful APIs in Python/Flask.
- Built a generic WebSocket service (Java + Redis pub/sub) to stream real-time status to other applications.
- Parallelized critical code paths to improve response times and overall reliability.

Education

M.Tech. in Computer Science and Engineering — Indian Institute of Technology Madras *2019 – 2022*

- Coursework: Advanced Data Structures and Algorithms, Advanced Programming Lab, Linear Algebra and Random Processes, Artificial Intelligence, Logic and Combinatorics for Computer Science

B.Tech in Computer Science and Engineering — JNTU Anantapur, Andhra Pradesh *2015 – 2019*

- Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks

Project Work

- Processing Algorithms for SPS of IRNSS (M.Tech Thesis), IIT Madras, May 2022 — Implemented processing algorithms for the Signal Processing Subsystem (SPS) of the Indian Regional Navigation Satellite System (IRNSS) under Prof. V. Kamakoti; built modules for ephemeris computation, ionospheric delay, and satellite position/clock-time corrections.

Certificates and Positions of Responsibility

- Udemy — Managing EC2 and VPC: AWS with Python and Boto3 Series
- Teaching Assistant, IIT Madras — Introduction to Programming (Jan–May 2021)
- Teaching Assistant, IIT Madras — Artificial Intelligence (Jul–Dec 2020)
- Teaching Assistant, IIT Madras — Natural Language Processing (Jan–May 2020)